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Department of Information Technology

Software Requirements Specification (SRS)

Project Title: SMART CAMPUS MANAGEMENT

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1.1 Problem Statement

Campus operations in many colleges such as resource booking, event management, student registrations, announcements, and approval workflows are still managed using traditional methods like registers, spreadsheets, emails, and messaging applications. These manual approaches make campus management inefficient and create difficulties in coordination between students, clubs, faculty members, and administrators. The absence of a centralized **Smart Campus Management Platform** leads to several operational challenges such as duplicate resource bookings, delayed approvals, scattered event information, difficulty in tracking student participation, and improper maintenance of administrative records. These problems exist due to the lack of a unified digital platform, manual resource allocation processes, absence of real-time conflict detection, and limited automation in notifications and reporting. Students, club leads, faculty coordinators, department HODs, administrators, and college management are affected by these issues as they face difficulties in accessing services, managing events, approving requests, maintaining records, and making effective decisions.

1.2 Need for the Project

The **Smart Campus Management Platform** is required to provide a centralized digital solution that automates and simplifies campus operations including resource scheduling, event management, student services, notifications, and administrative workflows. The system aims to reduce manual efforts, improve communication among stakeholders, prevent resource conflicts, and provide efficient management of campus activities through secure authentication, REST API communication, database transaction management, and automated notification services.

The existing campus management system faces several challenges such as manual booking of classrooms, laboratories, auditoriums, and other resources, lack of real-time availability checking, delayed approval processes, difficulty in maintaining event participation records, and absence of automated alerts. The motivation behind developing this platform is to create a smart and efficient campus ecosystem where students, faculty members, clubs, and administrators can interact through a single integrated system. The platform focuses on improving resource utilization, increasing transparency, reducing repetitive administrative tasks, and ensuring reliable operations using concurrency control and transaction management techniques.

1.3 Objectives



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The primary objective of **CampusOS** is to develop a centralized web-based platform that streamlines campus venue management, event organization, student participation, and administrative activities.

The specific objectives of the system are:

1. To reduce venue booking conflicts by implementing an automated scheduling system that prevents double booking of classrooms, laboratories, and auditoriums.
2. To provide a centralized event management platform where students can easily discover, search, and register for campus events.
3. To improve student engagement through seat reservation (RSVP), category-based event recommendations, and automated notifications.
4. To enhance administrative efficiency by implementing Role-Based Access Control (RBAC), event approval workflows, and club management features.
5. To ensure secure user authentication using JWT-based authentication and Role-Based Access Control (RBAC) for Students, Club Leads, and Administrators.

1.4 Scope

1.4.1 In Scope (What the System Will Do)

1. Manage venue booking requests for classrooms, laboratories, and auditoriums.
2. Prevent duplicate venue bookings through automated conflict detection.
3. Allow administrators to approve or reject venue booking requests.
4. Publish approved events on the student event portal.
5. Enable students to browse, search, filter, and register for campus events.
6. Implement seat reservation (RSVP) with event capacity validation to prevent overbooking.
7. Recommend similar events based on event categories.
8. Send notifications for new events, registration confirmations, event cancellations, and reminders before scheduled events.
9. Provide secure login using JWT authentication and Role-Based Access Control (RBAC).

1.4.2 Out of Scope (What the System Will Not Do)

1. Online payment or ticket purchasing for events.
2. Integration with external calendar platforms (e.g., Google Calendar or Microsoft Outlook).



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3. Live streaming of campus events.
4. AI-based event recommendations or chatbot assistance (recommendations will be category-based only).
5. Social networking features such as chat, comments, discussion forums, or messaging.
6. Attendance tracking using biometric devices, RFID, or facial recognition.
7. Native mobile application development (the initial version will be web-based only).
8. Integration with external College ERP or Learning Management Systems (LMS).

2. Problem Identification and Scope

2.1 Existing System

Many educational institutions continue to manage campus events, venue bookings, student participation, and administrative activities through traditional methods such as paper forms, spreadsheets, emails, notice boards, Google Forms, and messaging applications. Since these systems operate independently, there is no centralized platform to coordinate event management, resource allocation, or communication among different stakeholders.

When a Club Head intends to organize an event, they must manually contact the respective Department Event Coordinator to check venue availability and obtain approval. The availability of classrooms, laboratories, seminar halls, and auditoriums is often maintained in separate registers or spreadsheets, making it difficult to determine whether a venue has already been reserved. This frequently results in scheduling conflicts, delayed approvals, and inefficient utilization of campus resources.

Similarly, students receive event announcements through multiple communication channels, causing important notifications to be missed. Administrative tasks such as monitoring club activities, generating reports, maintaining records, and tracking user actions are performed manually, increasing workload and reducing overall operational efficiency.

As the number of students, clubs, and events increases, the existing system becomes increasingly difficult to manage due to the absence of automation, centralized control, and secure access mechanisms.

2.2 Current Process

The current campus event management process generally follows these steps:



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1. A Club Head plans an event for the department or student club.
2. The Club Head contacts the Department Event Coordinator through email, phone, WhatsApp, or paper forms to request a venue.
3. The Department Event Coordinator manually verifies the availability of the requested venue.
4. If the venue is available, the booking is approved manually.
5. Event information is shared through notice boards, WhatsApp groups, emails, or social media.
6. Students register using Google Forms or manual registration methods.
7. Administrative records and reports are maintained using spreadsheets or paper documents.

Since there is no centralized system, departments maintain their own records, making it difficult to monitor campus-wide activities and resource utilization.

2.3 Limitations of the Existing System

The current system suffers from several limitations that affect the efficiency of campus event management.

- Venue booking is performed manually, increasing the possibility of scheduling conflicts.
- There is no centralized database for managing campus resources and events.
- Real-time venue availability cannot be verified.
- Event announcements are distributed across multiple platforms, causing students to miss important updates.
- The approval process depends on manual communication, resulting in delays.
- Student registrations are not integrated with the event management system.
- Administrative reports require significant manual effort to prepare.
- No centralized audit mechanism exists to monitor administrative activities.
- Limited authentication and authorization increase security risks.
- Manual record maintenance increases the chances of human error.
- The existing process becomes inefficient during large college festivals and multiple simultaneous events.

2.4 Problems Identified

Based on the analysis of the current system, the following problems have been identified:



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- Duplicate booking of classrooms, laboratories, seminar halls, and auditoriums.
- Lack of centralized campus event management.
- Slow venue approval process.
- Poor coordination among Club Heads, Department Event Coordinators, and administrators.
- Students frequently miss important event notifications.
- Difficulty monitoring venue availability and booking history.
- Manual generation of reports and statistics.
- High administrative workload due to paper-based processes.
- Lack of transparency in booking and approval workflows.
- Inadequate security and access control.
- Absence of audit logs for monitoring administrative activities.

2.5 Proposed System

To overcome the limitations of the existing system, **CampusOS** is proposed as a centralized, web-based Smart Campus Management Platform that integrates venue booking, event management, student participation, authentication, notifications, and administrative governance into a single application.

The system provides secure Role-Based Access Control (RBAC), allowing different users to access only the features relevant to their responsibilities.

The system consists of the following user roles:

Principal & HOD (Super Admin)

The Principal and HOD act as the highest authority in the system and are responsible for:

- Managing the entire CampusOS platform.
- Creating and managing Department Event Coordinators.
- Managing departments and clubs.
- Monitoring campus-wide activities.
- Viewing reports and analytics.
- Monitoring audit logs.
- Sending campus-wide announcements.
- Managing system configurations and user permissions.

Department Event Coordinator



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Each department has a Department Event Coordinator responsible for:

- Reviewing venue booking requests submitted by Club Heads.
- Verifying venue availability.
- Approving booking requests.
- Monitoring department events.
- Viewing department-specific reports.

Club Head

Club Heads are responsible for organizing events within their clubs.

They can:

- Create new events.
- Select the required building, floor, and venue.
- Choose available dates and time slots.
- Submit venue booking requests.
- Manage club members.
- Track booking status.

Every booking request remains in the **Pending Approval** state until it is approved by the respective Department Event Coordinator.

Club Members

Club Members assist Club Heads in organizing and managing events.

They can:

- View assigned events.
- Assist in event preparation.
- Update event information (with permission).
- Coordinate event execution.

Students / Participants

Students and participants can:



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- Browse campus events.
- Search and filter events.
- Register for events.
- Reserve seats using RSVP.
- Receive event notifications.
- View registration history.

2.6 Advantages of the Proposed System

The proposed system offers several benefits over the existing manual process.

- Centralized management of campus events and resources.
- Automated venue booking with real-time availability checking.
- Prevention of duplicate bookings through conflict detection.
- Structured approval workflow for venue requests.
- Secure authentication using JWT and Role-Based Access Control (RBAC).
- Online event registration with seat reservation.
- Automated notifications and reminders.
- Dashboard-based monitoring for administrators.
- Automated report generation and analytics.
- Audit logging for improved accountability and security.
- Better communication among students, clubs, coordinators, and administrators.

2.7 Innovation

CampusOS introduces several innovative features that improve the efficiency of campus management.

- Centralized Smart Campus Management Platform.
- Secure Role-Based Access Control (RBAC).
- JWT-based authentication.
- Automated venue conflict detection.
- Interactive venue selection based on Building → Floor → Venue hierarchy.
- Online RSVP-based event registration.
- Real-time notification service.
- Dashboard analytics for monitoring campus activities.
- Comprehensive audit logging.



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- Scalable architecture capable of supporting future AI-powered features such as intelligent event recommendations and chatbot assistance.

2.8 Expected Benefits

Time Saving

- Faster venue booking process.
 - Reduced approval time.
 - Instant event notifications.
 - Quick student registration.
 - Automated report generation.
-

Cost Reduction

- Reduced paper usage.
- Lower printing costs.
- Reduced administrative workload.
- Efficient utilization of campus resources.

Accuracy

- Prevents duplicate venue bookings.
- Reduces manual errors.
- Improves data consistency.
- Generates accurate reports.
- Maintains complete audit records.

User Convenience

Principal & HOD

- Centralized monitoring of all campus activities.
- Access to reports and analytics.
- Complete user and system management.

Department Event Coordinator



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- Easy approval of venue bookings.
- Department-level event monitoring.
- Department reports and analytics.

Club Head

- Simple event creation and venue booking.
- Easy booking status tracking.
- Efficient club management.

Club Members

- Better coordination of event activities.
- Task management support.
- Easy access to assigned responsibilities.

Students / Participants

- Easy event discovery.
- Online registration and seat reservation.
- Real-time notifications.
- Convenient access to registration history.

3. Feasibility Study

3.1 Technical Feasibility

- **Technology Stack:** React.js (Vite) [Frontend], Node.js + Express.js [Backend REST APIs], PostgreSQL / MySQL [Relational Database].
- **Software Tools:** VS Code, Git/GitHub Monorepo, Postman, MySQL Workbench / pgAdmin.
- **Hardware Requirements:** Standard development machines (8 GB RAM, Intel i5/AMD Ryzen 5, 256 GB SSD) and cloud hosting instances.

3.2 Economic Feasibility

- **Estimated Cost:** ₹0 (Zero software licensing cost).
- **Open-Source Software Used:** Entirely built using open-source technologies (Node.js, Express, React, PostgreSQL/MySQL, Git).



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3.3 Operational Feasibility

- The platform features an intuitive, responsive Single Page Application (SPA) dashboard.
- Role-based dashboards guide users based on their access permissions (Student, Club Lead, Coordinator, Admin).

3.4 Project Scope & Deliverables

- **Major Modules:**
 - Module 1: Authentication & RBAC (JWT, Gmail OAuth).
 - Module 2: Campus Resources & Scheduling Engine (Concurrency Lock & Buffer Validation).
 - Module 3: Student Services & Seat RSVP Engine (Backout Handling).
 - Module 4: Multi-Channel Notification Engine (Scheduled Workers).
 - Module 5: Admin Governance, Audit Trail & Analytics Exporter.
- **Deliverables:** Production codebase (GitHub Monorepo), Database SQL DDL scripts, SRS Document, Test Reports, and System Documentation.

4.1 Stakeholder Analysis

Stakeholder	Role	Expectations
Admin	Manage system and oversee role permissions. And Decision making regarding overall campus resources and infrastructure.	Easy management and robust database concurrency. Accurate analytics and quickly exportable metrics for venue usage and attendance.
User(Student)	Use application to browse the discovery feed and reserve event seats.	Fast & reliable RSVP process and accurate multi-channel reminders.
Club Lead/Faculty(Coordinator)	Schedule venues and organize large-scale campus events and Monitor club activities	Seamless time-slot selection with a conflict prevention engine ensuring zero double-bookings. And Report Analysis



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5. Functional Requirements

- **FR1 (Multi-Provider User Registration):** The system shall allow internal students and faculty to register using institutional credentials, while external participants shall be permitted to register and log in via Gmail OAuth 2.0.
- **FR2 (Login Authentication & Token Issuance):** The system shall authenticate users using secure login credentials and issue signed JSON Web Tokens (JWT) for stateless session authorization.
- **FR3 (Role-Based Access Control - RBAC):** System middleware shall enforce RBAC permissions across **five hierarchical roles**: Principal & HOD (Super Admin), Department Event Coordinator (Faculty Admin), Club Head / President, Club Member, and Student / Participant.
- **FR4 (Session Management):** The system shall manage user sessions securely using JWT tokens and automatically expire inactive sessions after a predefined duration.
- **FR5 (User Profile Management):** The system shall allow users to view and update their profile details, including full name, contact information, department, and profile image.
- **FR6 (Venue Directory & Search):** The system shall maintain a searchable catalogue of campus venues (auditoriums, seminar halls, labs) filterable by building, floor, capacity, location, and available equipment.
- **FR7 (Time-Slot Picker UI):** The system shall provide an interactive calendar interface allowing Club Heads and Faculty to select date and time windows for venue reservations.
- **FR8 (Interval Overlap Detection Algorithm):** The backend shall enforce non-overlapping time logic for all booking requests using the formula:
$$((Start_{{\text{new}}} < End_{{\text{existing}}} + Buffer) \wedge (End_{{\text{new}}} + Buffer > Start_{{\text{existing}}}))$$
- **FR9 (Venue Buffer Time Validation):** The scheduling engine shall automatically enforce a configurable setup/teardown buffer time (e.g., 15 minutes) between back-to-back event bookings for the same venue.
- **FR10 (Pessimistic Concurrency Locking):** Booking transactions shall execute within SQL transaction blocks utilizing row-level pessimistic locking (SELECT FOR UPDATE) to guarantee **zero duplicate venue bookings** under simultaneous concurrent load.
- **FR11 (Club Management):** Administrators shall be able to create, update, disable, and manage campus clubs. Club Heads shall be authorized to manage their specific club details and organizing team members.
- **FR12 (Multi-Tier Approval Routing & Dynamic Slot Lifecycle):** The system shall enforce a dynamic approval and visual slot-status lifecycle:



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- **Pending State (Yellow):** When a Club President submits a venue request, the slot status on the calendar visually transitions to **Yellow ("Approval Pending")** and routes to the Faculty Coordinator's inbox.
- **Competitive Pending Requests:** While a slot is in the pending state (Yellow), other clubs shall still be permitted to submit booking requests for that same time window.
- **First-Approved Priority (Red):** Whichever booking request receives faculty approval **first** shall secure the slot. Upon approval, the venue status instantly turns **Red ("Booked")**, locking the slot and automatically rejecting competing pending requests for that window.
- **College-Level Event Routing:** Official college-level event proposals shall route to the **HOD / Principal** for final administrative sign-off.
- **Direct Faculty Booking:** Faculty Coordinators shall possess direct booking privileges for official department events, which instantly locks the room as **Red ("Booked")** without requiring a pending step.
- **FR13 (Approval Inbox & Decision Logging):** Approvers shall review pending booking requests and issue One-Click **Approve**, **Reject**, or **Request Modification** decisions. Rejections shall require a mandatory recorded reason entry.
- **FR14 (Event Discovery Feed, Search & Filtering):** The system shall provide a student event discovery feed with options to search and filter events by category (technical, cultural, sports), organizing club, venue, date, and status.
- **FR15 (Seat Reservation / RSVP & Eligibility Verification):** Students shall reserve seats up to the venue's allocated capacity limit. The system shall verify eligibility criteria (e.g., department, academic year) prior to confirming registrations.
- **FR16 (Participant Backout & Seat Recovery):** If a registered student cancels their event RSVP, the system shall atomically decrement the booked seat count, restoring seat availability for other participants.
- **FR17 (Category-Based Event Recommendations):** The student portal shall display recommended upcoming events based on the user's previous registration history and interest categories.
- **FR18 (Role-Specific Dashboards):** The system shall render dedicated, role-tailored dashboards for Students, Club Leads, Faculty Coordinators, and Super Admins displaying relevant metrics, actions, and schedules.
- **FR19 (Multi-Channel Notification System):** The system shall send real-time broadcast alerts upon event publishing/status changes and deliver automated background reminders **2 days** and **2 hours** prior to event start times.



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- **FR20 (System Audit Logging):** All administrative overrides, user logins, role modifications, event approvals, and venue decisions shall be logged in an immutable AdminLogs database table.
- **FR21 (Analytics & Report Exporter):** Administrators shall generate and export venue utilization, club activity, and student attendance metrics in PDF and CSV/Excel formats.

6. Non-Functional Requirements

Performance

- System response time shall be less than **3 seconds** under normal load.
- Dashboard pages shall load within **2 seconds**.
- The system shall support simultaneous access by at least **500 users**.
- Database queries shall be optimized for quick event retrieval.

Security

- JWT-based authentication shall be implemented.
- Passwords shall be encrypted using **bcrypt**.
- Role-Based Access Control (RBAC) shall restrict unauthorized access.
- All user inputs shall be validated on both client and server sides.
- Login attempts shall be protected using rate limiting.

Reliability

- System availability shall be at least **99%** during operational hours.
- User data shall be stored reliably in the database.
- Failed transactions shall not corrupt existing data.
- Event bookings shall remain consistent even during concurrent requests.

Scalability

- The system shall support up to **1000 registered users**.
- It shall allow future addition of new roles without major architectural changes.
- The system architecture shall support future modules such as payment integration and certificate generation.



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Usability

- The interface shall be simple and intuitive.
- Navigation shall be consistent across all dashboards.
- Responsive design shall support desktops, tablets, and mobile devices.
- Users shall be able to complete common tasks within a few clicks.

Maintainability

- The application shall follow a modular architecture.
- Backend code shall be organized into controllers, routes, middleware, and services.
- APIs shall follow RESTful standards.
- Source code shall be documented and version-controlled using Git.

Portability

The application shall run on:

- Windows
- Linux
- macOS

The web application shall be accessible through modern browsers including:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

Compatibility

The system shall support:

- Postgresql
- Node.js 20+
- Express.js
- Modern web browsers

Availability



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- The system shall remain operational throughout college working hours.
- Scheduled maintenance shall not affect stored data.

7. System Requirements

Hardware Requirements

Development Environment

- Intel Core i5 (10th Generation or above) / AMD Ryzen 5 or equivalent
- Minimum 8 GB RAM (16 GB recommended)
- 256 GB SSD (512 GB recommended)
- Stable internet connection

Server (Deployment)

- Quad-Core Processor
- 8 GB RAM or higher
- 100 GB Storage
- Broadband Internet Connection

Software Requirements

Operating System

- Windows 10/11
- Ubuntu Linux 22.04 or later
- macOS (optional)

Backend

- Node.js (v20 or later)
- Express.js

Frontend

- React

Database



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- Postgresql

Authentication & Security

- JWT (jsonwebtoken)
- bcrypt
- CORS
- express-rate-limit
- express-validator

Development Tools

- Visual Studio Code
- Git
- GitHub
- Postman (API Testing)

Browser Support

- Google Chrome (Recommended)
- Microsoft Edge
- Mozilla Firefox
- Safari

8. Use Case Diagram

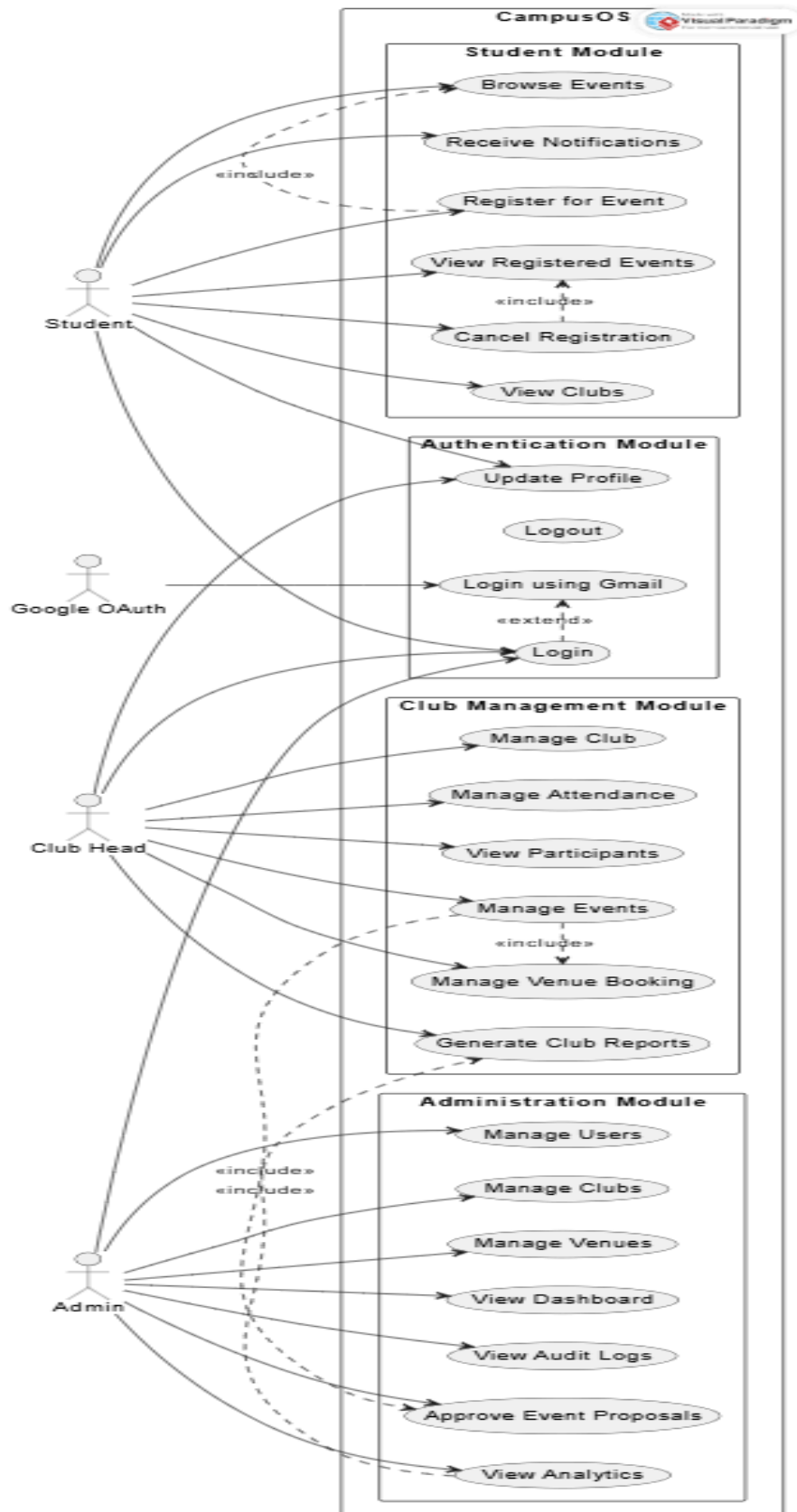


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9. Use Case Descriptions

The Use Case Diagram represents the functional interactions between the actors and the CampusOS – Smart Campus Club & Event Management System. It identifies the primary actors, including Student, Club Head, Admin, and the external Google OAuth service, and illustrates their interactions with the various system modules.

The Student can authenticate using either standard login credentials or Google Sign-In, update their profile, browse clubs and events, register for events, cancel event registrations, view registered events, and receive notifications related to club activities and upcoming events. The system ensures that students can only register for available events and maintains accurate records of their registrations.

The Club Head is responsible for managing club activities, including maintaining club information, organizing and managing events, handling venue booking requests, managing attendance, viewing participant details, and generating club reports. Event proposals and venue booking requests are submitted through the system to ensure proper approval before an event is conducted.

The Admin manages the overall operation of the system by maintaining user accounts, clubs, and venues, approving event proposals, monitoring system activities through the dashboard, viewing analytics, and accessing audit logs. The administrator ensures that all club activities comply with institutional policies and that resources such as venues are allocated efficiently.

The Google OAuth service acts as an external authentication provider, enabling users to securely log in using their Google accounts. This integration simplifies the authentication process while maintaining secure access to the system.

Overall, the Use Case Diagram provides a high-level overview of the functional requirements of the CampusOS and illustrates how users and the external authentication service interact with different system modules to support efficient club management, event organization, venue booking, and student participation across the campus.

10. Data Requirements

10.1 The CampusOS uses the following database tables:

- **Roles** – Defines user roles and access permissions within the system.



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- **Users** – Stores user account details, authentication information, and assigned roles.
- **Venues** – Maintains information about campus venues, including their capacity and location.
- **Clubs** – Stores details of student clubs and their respective club heads.
- **Events** – Stores event details such as title, schedule, organizer, and event status.
- **Bookings** – Stores venue booking records, including the selected venue, event, booking date, and booking status (**Available, Pending, or Booked**).
- **Attendance** – Records student attendance for conducted events.
- **Notifications** – Stores notifications and event-related updates sent to users.
- **AdminLogs** – Maintains a record of important administrative activities for auditing and system monitoring.

10.2 Important Attributes

TABLE	IMPORTANT ATTRIBUTES
Roles	role_id, role_name
Users	user_id, full_name, email, password_hash, department, role_id
Venues	venue_id, venue_name, capacity, location
Clubs	club_id, club_name, description, club_head_id
Events	event_id, club_id, created_by, title, description, event_date, start_time, end_time, status
Bookings	booking_id, event_id, venue_id, requested_by, booking_date, status
Attendance	attendance_id, event_id, student_id, status
Notifications	notification_id, user_id, event_id, title, message, created_at
AdminLogs	log_id, admin_id, booking_id, action, created_at

10.3 Relationships

- One **Role** can be assigned to many **Users**.
- One **User** can manage one or more **Clubs** as a Club Head.



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- One **Club** can organize multiple **Events**.
- One **User** can create multiple **Events**.
- One **Venue** can have multiple **Bookings** on different dates and time slots.
- One **Event** is associated with one **Booking** for venue reservation.
- One **Event** can have multiple **Attendance** records.
- One **Student (User)** can have attendance records for multiple **Events**.
- One **User** can receive multiple **Notifications**.
- One **Event** can generate notifications for multiple **Users**.
- One **Admin (User)** can perform multiple administrative actions, which are recorded in **AdminLogs**

11. Assumptions and Constraints

11.1 Assumptions

- **A1. Internet Connectivity:** Users have access to a stable internet connection to use the web application.
- **A2. User Authentication:** Every user (Student, Club Head, Faculty, HOD, Principal, Admin) has valid login credentials provided by the institution.
- **A3. Device Availability:** Users can access the system using desktops, laptops, with a modern web browser.
- **A4. Authorized Users:** Only authorized users can access system features according to their assigned roles and permissions.
- **A5. Venue Information:** All venue details such as capacity, location, facilities, and availability are maintained accurately by the authorized faculty.
- **A7. Admin Availability:** The Admin (Principal, HoDs), Faculty Coordinator, or Department Coordinator regularly reviews and approves or rejects venue booking requests.
- **A8. Database Availability:** The database server remains available and stores updated information about users, venues, events, and bookings.
- **A9. Notification Service:** The notification system is available to send in-app and email notifications to users.
- **A10. Concurrent Users:** The system supports multiple users accessing the application simultaneously while maintaining data consistency.

11.2 Constraints



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- **C1. Limited Development Time:** The project must be completed within the academic semester (approximately 12 weeks).
- **C3. Internet Dependency:** The application requires an active internet connection and does not provide complete offline functionality.
- **C4. Browser Compatibility:** The system is optimized for modern web browsers such as Google Chrome, Microsoft Edge.
- **C6. Security Policies:** Access to the system is controlled through authentication and Role-Based Access Control (RBAC), preventing unauthorized access.
- **C7. Venue Availability:** Venue bookings are allowed only for venues available in the campus database and during permitted operating hours.
- **C9. Concurrent Booking Restriction:** Only one booking is allowed for a specific venue and time slot. Simultaneous booking requests are managed using database transactions and row-level locking.

12. Future Enhancements

- **AI Assistant:** Integration of a RAG-based AI assistant for answering campus club policy queries.
- **QR Code Check-ins:** Automated QR code ticket scanning at venue entry doors for live attendance verification.
- **Mobile Application:** Cross-platform React Native mobile application for native mobile push notifications.

13. References

1. Node.js & Express.js Official Documentation — <https://nodejs.org> / <https://expressjs.com>
2. React.js (Vite) Documentation — <https://vitejs.dev>
3. PostgreSQL ACID Concurrency & Row Locking — <https://www.postgresql.org/docs/current/explicit-locking.html>
4. IEEE Standard 830-1998 for Software Requirements Specifications.